Nathaniel Kenschaft

nkenscha@andrew.cmu.edu | linkedin.com/in/nkenschaft | github.com/nkenschaft

Education

Carnegie Mellon University *Pittsburgh, PA* May 2023 - PRESENT
B.S. in Mathematical Sciences GPA 3.7/4.0
Pursuing advanced coursework in mathematics, as well as adjacent fields. Planning to pursue PhD in math upon graduation. Will graduate in May 2027.

Thomas Jefferson High School for Science and Technology *Alexandria, VA* June 2019 - June 2023
High School Diploma GPA 4.0/4.0
Activities include machine learning club, head leadership position in school systems administrators (~2300 users), and high leadership positions in marching band.

Selected Coursework

Mathematics	algebraic and point-set topology, real and functional analysis, measure theory, category theory, probability modeling
Computer Science	discrete differential geometry, machine learning, artificial intelligence
Miscellaneous	philosophy and history of science, biological foundations of behavior
Kaggle Certificates	deep learning, computer vision, machine learning

Research and Work Experience

Summer Experiences in Mathematical Sciences *Pittsburgh, PA* Summer 2025
Worked with Professor Tomasz Tkocz and Postdoctoral Associate Dr. Martin Rapaport on an open problem in geometry and measure theory involving the volume of repeated Minkowski sums and averages. Produced an exposition of existing research and proposed a new avenue of exploration for the problem.

Machine Learning Intern at Verseon Corporation *Princeton, NJ* Summer 2024
Built a novel drug generation system using deep reinforcement learning and sequence generation models.

Capstone Project August 2022 - June 2023
High school capstone research project using reinforcement learning (PPO+LSTM) to train agents to gather food in a simulated environment. Used model to compare how humans interact on a larger scale when presented with limited resources. Wrote an associated paper to report findings.

Intern at NOVA Community College Summer Camp *Annandale, VA* Summer 2022
Student employee at a summer camp for upper elementary through middle-school-age children. Taught students to build circuits and program Arduino microcontrollers.

Volunteer at Codefy *Virtual* Spring 2021, Summer 2022
Helped teach a course on basic Java programming, mentoring individual students on coursework. Later planned and taught an advanced Python programming course.

Skills

Professional	LaTeX, Google Sheets/Excel
Machine Learning	deep learning (ANN, CNN, LSTM); reinforcement learning (PPO, deep RL, multi-agent); statistical (kNN, k-means, SVMs, decision trees, random forests)
Artificial Intelligence	graph search, game modeling & strategy, natural language processing, genetic algorithms
Programming	Python, C, Java, Rust
Linux	standard distros, command line tools, system installation & configuration